Natalie McDonald

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EDUCATION

Virginia Tech, Blacksburg, VA
B.S. in Mechanical Engineering, Robotics and Mechatronics Concentration — Expected Spring 2027
Cumulative GPA: 3.13, In Major GPA: 3.44

SKILLS

MATLAB • Simulink • Simscape • Python • SolidWorks • Siemens NX • Autodesk Fusion • AutoCAD • Revit • C/C++ • Mathematica • FDM 3D Printing

WORK EXPERIENCE

Mechanical Engineering Intern, Noli Robotics
Remote | Aug 2026 – Present

- Design and modify CAD models for aerial and ground units, including housing and mechanical assemblies.
- Operate resin and FDM 3D printers and test printed parts for fit, assembly, and tolerances while helping reduce unit costs.

Mechanical Engineering Intern, M.C. Dean
Tysons, VA | June 2026 – Present

- Developed architectural and mechanical design documentation for mission-critical modular data center infrastructure, producing engineering drawings in Revit.
- Designed HVAC systems for industrial facilities using flow analysis; coordinated piping layouts to support integrated building systems.
- Create and revise AutoCAD one-line diagrams for industrial building systems to support project bid proposals.

Undergraduate Research Assistant, Virginia Tech Nonlinear Systems Lab
Blacksburg, VA | Jan 2026 – Present

- Designed and fabricated a planar double pendulum test platform to experimentally validate a high-gain observer for nonlinear disturbance estimation.
- Authored technical documentation detailing equations of motion derivations, analytical results, and CAD assembly models supporting ongoing controls research.

Mechanical Engineering Undergraduate Teaching Assistant, Virginia Tech
Blacksburg, VA | Aug 2025 – Present

- Mentor 40+ students in MATLAB- and Python-based numerical methods through office hours and individualized problem-solving guidance.

Engineering Intern, Army DEVCOM Chemical Biological Center
Edgewood, MD | June 2025 – Sept 2025

- Designed and executed an experimental test plan to evaluate additively manufactured materials subjected to repeated chemical decontamination cycles for military sustainment applications.
- Documented experimental results and collaborated with engineers to support material selection and future procurement decisions.

Undergraduate Research Assistant, GrayUR Interdisciplinary Research Group at Virginia Tech
Blacksburg, VA | Sept 2024 – Aug 2025

- Collaborated on the design of a low-cost fixed-wing UAV capable of carrying a multispectral imaging payload.
- Supported team organization, fuselage design, and material selection to improve project operations and performance.

CAMPUS INVOLVEMENT

Outreach Chair, Virginia Tech American Society of Mechanical Engineers (ASME)

Blacksburg, VA | June 2026 – Present

- Manage chapter communications and social media to increase student engagement
- Develop industry partnerships, expand networking, sponsorships, and professional development opportunities for members.

Finance Lead, Virgina Tech 3D Printed Aircraft Competition (3DPAC)

Blacksburg, VA | Feb 2025 – Present

- Manage team finances by developing budgets, tracking expenditures, and securing sponsorships to support competition activities.
- Lead outreach initiatives by managing the team's social media presence and organizing events to recruit members and engage industry sponsors.
- Manufacture flight-ready aircraft components via additive manufacturing, optimizing CAD models and print parameters to improve structural performance and weight efficiency.

Propulsion Sub-Team Member, MachWorks Design Team at Virginia Tech

Blacksburg, VA | June 2024 – Jan 2026

- Developed a dynamic propulsion model in MATLAB, Simulink, and Python to characterize the relationship between throttle position and engine thrust.
- Validated propulsion models to support future engine testing and system performance analysis.

INTERESTS

CAD Design and 3D printing • Programming • Hiking • American Sign Language • Painting • Strength Training